Stochastic optimization

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Introduction

Consider the general deterministic program

$$\begin{aligned} &\min g_0(x) \\ &\text{s.t. } g_i(x) \leq 0, i = 1, \ldots, m \\ &\quad x \in X \subset \mathcal{R}^n. \end{aligned}$$

All the parameters are assumed to be perfectly known. Realistic?

- measurement errors;
- uncertainties on the future;
- data unavailable;
- ...

Mathematical programming and stochastic programming

- Mathematical programming (optimization): typically: decision problem (where the meaning of the term “decision” is broad).
- Stochastic programming concerns decision under uncertainty, the uncertainty being represented by means of random parameters.

$$\begin{aligned} &\text{“}\min_{x \in X}\text{”} g_0(x, \xi) \\ &\text{s.t. } g_i(x, \xi) \leq 0, \ i = 1, \ldots, m, \end{aligned}$$

where ξ is a random vector. Meaning of “min”?

Assumption: we can represent the uncertainty by means of the (joint) probability distribution.

The farmer problem

Source: Birge and Louveaux, Section 1.1.

Scenarios approach

- Assumption: finite random vector. A realization = a scenario.
- Even if the random vector is continuous, a discrete approximation is often useful.

- A European farmer has 500 acres of land.
- He cultivates wheat, corn and sugar beets.
- Livestock food: at least 200T of wheat and 240T of corn.
- Must buy if less production, but can sell the overproduction.
- The purchase cost is 40% greater than the sale cost.
- The farmer can sell the sugar beets at \$36/T for the first 6000 tons, and \$10/T thereafter, due to European quotas.

The farmer problem II

Culture	Wheat	Corn	Sugar beets
Average yield (T/acre)	2.5	3	20
Planting cost ($/acre)	150	230	260
Selling price ($/T)	170	150	36 ($\leq$ 6000T), 10
Buying price ($/T)	238	210	-
Minimum required (T)	200	240	-

Notations:

- x_1, x_2, x_3: acres for wheat, corn, sugar beets;
- y_1, y_2: tons of bought wheat and corn;
- w_1, w_2: tons of sold wheat and corn;
- w_3, w_4: tons of sold sugar beets, at high price and at low price.

How to decide the surface to allocate to each plant?

The farmer problem: deterministic version

Linear program:

$$
\begin{aligned}
\min\ & 150x_1 + 230x_2 + 260x_3 + \\
& 238y_1 - 170w_1 + 210y_2 - 150w_2 - 36w_3 - 10w_4 \\
\text{s.t. } & x_1 + x_2 + x_3 \leq 500; \\
& 2.5x_1 + y_1 - w_1 \geq 200; \\
& 3x_2 + y_2 - w_2 \geq 240; \\
& w_3 + w_4 \leq 20x_3; \\
& w_3 \leq 6000; \\
& x_1, x_2, x_3, y_1, y_2, w_1, w_2, w_3, w_4 \geq 0.
\end{aligned}
$$

The farmer problem: deterministic solution

Total (expected) profit: $118600. Details:

Culture	Wheat	Corn	Sugar beets
Surface (acres)	120	80	300
Production (T)	300	240	6000
Sales (T)	100	-	6000
Purchase (T)	-	-	-

Production can however increase or decrease by 20%, depending on the weather.
Simplified setting:

- good year (for every plant, the production is 20% higher),
- average year,
- bad year (for every plant, the production is 20% lower).

The prices do not change.

The farmer problem: scenario solutions

New optimal solutions?

Good year. Total profit: $167667.

Culture	Wheat	Corn	Sugar beets
Surface (acres)	183.33	66.67	250
Production (T)	550	240	6000
Sales (T)	350	-	6000
Purchases (T)	-	-	-

Bad year. Total profit: $59950.

Culture	Wheat	Corn	Sugar beets
Surface (acres)	100	25	375
Production (T)	200	60	6000
Sales (T)	-	-	6000
Purchases (T)	-	180	-

The farmer problem: scenarios

The decisions considerably change with the weather conditions, but how to know them when deciding what to plant?
The decisions (x_1, x_2, x_3) have to be made now, but sales and purchases $(w_i, i = 1, \ldots, 4, y_j, j = 1, 2)$ depend on yields.

Scenarios: $s \in \{1, 2, 3\}$.

- $s = 1$ yields higher than average,
- $s = 2$ equal to average,
- $s = 3$ lower than average.

New variables: w_{is} and y_{is}.

The farmer problem: extensive form

- We now want to maximize the expected profit.
- Assumption: 3 equiprobable scenarios.

$$
\begin{aligned}
\min\ & 150x_1 + 230x_2 + 260x_3 + \\
& + \sum_{s=1}^{3} \frac{1}{3}(238y_{1s} - 170w_{1s} + 210y_{2s} - 150w_{2s} - 36w_{3s} - 10w_{4s}) \\
\text{s.t. } & x_1 + x_2 + x_3 \leq 500; \\
& 3x_1 + y_{11} - w_{11} \geq 200; 2.5x_1 + y_{12} - w_{12} \geq 200; \\
& 2x_1 + y_{13} - w_{13} \geq 200; \\
& 3.6x_2 + y_{21} - w_{21} \geq 240; 3x_2 + y_{22} - w_{22} \geq 240; \\
& 2.4x_2 + y_{23} - w_{23} \geq 240; \\
& w_{31} + w_{41} \leq 24x_3; w_{32} + w_{42} \leq 20x_3; w_{33} + w_{43} \leq 16x_3; \\
& w_{31} \leq 6000; w_{32} \leq 6000; w_{33} \leq 6000; \\
& x, y, w \geq 0.
\end{aligned}
$$

$\rightarrow$ extensive form (or deterministic equivalent).

The farmer problem: stages

- Seeding decisions: first-stage decisions;
- Sale and purchase decisions: second-stage decisions.

Total profit: \$108390.

	Culture	Wheat	Corn	Sugar beets
First stage	Surface (acres)	170	80	250
$s = 1$	Production (T)	510	288	6000
	Sales (T)	310	48	6000
	Purchases (T)	-	-	-
$s = 2$	Production (T)	425	240	5000
	Sales (T)	225	-	5000
	Purchases (T)	-	-	-
$s = 3$	Production (T)	340	192	4000
	Sales (T)	140	-	4000
	Purchases (T)	-	48	-

Observations

- The optimal decision has changed!!!
- Decision under perfect information: if the farmer could know the scenario in advance, or wait to observe the realization of the random variables (r.v.) (wait-and-see approach), the average annual profit would be $115406.
- The difference with the optimal decision under uncertainty is called expected value of perfect information (EVPI): profit loss due to uncertainty (here, $7016).

Observations: value of the stochastic solution

- Replacing ξ by $\mathbb{E}[\xi]$ in the problem leads to the expected value (EV) problem, delivering the expected value solution.
- The expectation of the expected value (EEV) problem is obtained by computing the expected profit over the scenarios when the EV solution is always used.
- Here, the expectation of scenarios is the average year, but in general, the expectation will not necessarily correspond to a pre-existent scenario.
- If the farmer only uses the average information, the average profit would be \$107240.
- It leads to a loss of \$1150 with respect to the solution of the stochastic problem. This difference is known as value of the stochastic solution (VSS).

Example: the newsvendor problem

Source: Birge and Louveaux, Section 1.1.

- A newsvendor has to decide how many newspapers to buy in order to maximize his profit. However he does not know in advance how many newspapers he will be able to sell during a day (the demand).
- Each newspaper costs c, and can be sold at a price q.
- The newsvendor can turn back the unsold newspapers at the end of the day, and obtain a price r for each of them
- Knowing the probability distribution $F(t) = P(\xi \leq t)$, how many newspapers should the newsvendor buy in order to maximize his revenue?

The newsvendor problem (cont'd)

- With the previous definitions, the newsvendor would like to solve the following optimization problem:

$$\max_{x \geq 0} -cx + \mathcal{Q}(x),$$

- $\mathcal{Q}(x)$ is the expected sale amount if the newsvendor buys x newspapers:

$$\mathcal{Q}(x) = \mathbb{E}_{\boldsymbol{\xi}}[Q(x, \boldsymbol{\xi})].$$

- Here $Q(x, \xi)$ is the amount of money obtained by the newsvendor if he buys x newspapers and the demand is ξ.

The newsvendor problem (cont'd)

- As previously, we could construct an equivalent linear problem (presented in Birge and Louveaux).
- Can we simplify? It is easy to see that

$$Q(x,\xi) = \begin{cases} qx & \text{if } x \le \xi, \\ q\xi + r(x-\xi) & \text{if } x \ge \xi. \end{cases}$$

Therefore,

$$\begin{aligned} \mathcal{Q}(x) = \mathbb{E}_{\boldsymbol{\xi}}[Q(x,\boldsymbol{\xi})] &= \int_{-\infty}^{\infty} Q(x,\xi)\, dF(\xi) \\ &= \int_{-\infty}^{x} (q\xi + r(x-\xi))\, dF(\xi) + \int_{x}^{\infty} qx\, dF(\xi). \end{aligned}$$

The newsvendor problem (cont'd)

Therefore, we have

$$\begin{aligned}\mathcal{Q}(x) &= (q-r)\int_{-\infty}^{x} \xi\, dF(\xi) + rx\int_{-\infty}^{x} dF(\xi) + qx\int_{x}^{\infty} dF(\xi) \\ &= (q-r)\int_{-\infty}^{x} \xi\, dF(\xi) + rxF(x) + qx(1-F(x)) \\ &= (q-r)\left[\int_{-\infty}^{x} \xi\, dF(\xi) - xF(x)\right] + qx.\end{aligned}$$

Integration by parts

Assume that F satisfies $\lim_{t\to-\infty} tF(t) = 0$.

We can then integrate by parts to obtain:

$$\begin{aligned}\int_{-\infty}^{x} \xi \, dF(\xi) &= \xi F(\xi)|_{-\infty}^{x} - \int_{-\infty}^{x} F(\xi)\, d\xi \\ &= xF(x) - \int_{-\infty}^{x} F(\xi)\, d\xi.\end{aligned}$$

Thus,

$$\mathcal{Q}(x) = qx - (q-r)\int_{-\infty}^{x} F(\xi)\, d\xi.$$

Solution of the second stage

Recall the initial problem...

$$\max_{x \geq 0} -cx + \mathcal{Q}(x),$$

We have to solve this problem. We will consider the associated optimality conditions.

Assuming $x > 0$, the optimal order quantity x^* is obtained by setting the derivative of the objective function to zero. As

$$\frac{d}{dx}\mathcal{Q}(x) = q - (q - r)F(x),$$

we have

$$-c + q - (q - r)F(x) = 0$$

The newsvendor problem (cont'd)

The solution x^* is therefore

$$x^* = F^{-1}\left(\frac{q-c}{q-r}\right).$$

Example: $c = 0.15$, $q = 0.25$, $r = 0.02$, $\xi \sim N(650, 80^2)$. Then

$$x^* = N^{-1}_{(650,80^2)}(0.1/0.23).$$

Since $\xi = 80Z + 650$ where $Z \sim N(0,1)$ has distribution function Φ, it is easy to show that

$$x^* = 80\Phi^{-1}(0.1/0.23) + 650 \approx 636.86.$$

In Julia, we can compute this value as

```
using Distributions
d = Normal(650,80)
quantile(d, 0.1/0.23)
```

Marginal revenue

Other interpretation, more intuitive: assume that the vendor has bought t newspapers. What is the expected marginal revenue if he buys an additional newspaper? From an economic point of view, we would like this marginal revenue to be equal to 0.

The expected marginal revenue (MR) is

$$\begin{aligned} MR(t) &= -c + qP[\xi \geq t] + rP[\xi \leq t] \\ &= -c + q(1 - F(t)) + rF(t). \end{aligned}$$

If we set the marginal revenue to 0, we get

$$MR(t) = 0 \text{ iff } F(t) = \frac{q-c}{q-r},$$

and we recover the previous solution.